

Computability: Recitation 3

April 6th, 2025

1 NFA example - L_{mix}

Given a regular language L , show that

$$L_{\text{mix}} := \{w \in \Sigma^* : \exists z \in \Sigma^{|w|} \text{ s.t. } z_1 w_1 \cdots z_{|w|} w_{|w|} \in L\} \in \text{NREG}.$$

Example $\Sigma = \{a, b\}$, $L = \{abaa\}$.

$w = ba \in L_{\text{mix}}$ as $z_1 = a, z_2 = a$ satisfy the constraints. However, $w = aa \notin L$ since no z_1, z_2 satisfy the constraints. In fact, $L_{\text{mix}} = \{ba\}$.

1.1 Solution

Let $\mathcal{A} = \langle Q, \Sigma, \delta, q_0, F \rangle$ be a DFA that decides L . Define an NFA $\mathcal{B} = \langle Q, \Sigma, \eta, \{q_0\}, F \rangle$ where:

$$\eta(q, \sigma) = \{\delta(\delta(q, \alpha), \sigma) : \alpha \in \Sigma\}$$

The idea is that when reading a letter $\sigma = w_i$, the NFA first “guesses” a letter $\alpha = z_i$ and goes to a state that the DFA would reach after reading $z_i w_i$.

We prove that $L(\mathcal{B}) = L_{\text{mix}}$ by containments:

- Let $w = w_1 \cdots w_n \in L_{\text{mix}}$. Then there exists $z = z_1 \cdots z_n$ such that $z_1 w_1 \cdots z_n w_n \in L$. Let $r_0, s_1, r_1, s_2, r_2, \dots, s_n, r_n$ be an accepting run of $\mathcal{A}$ on $z_1 w_1 \cdots z_n w_n$. We claim that $r_0, r_1, \dots, r_n$ is an accepting run of $\mathcal{B}$ on w . Indeed, $r_0 \in \{q_0\}$ and $r_n \in F$, and for every $0 \leq i < n$ we have:

$$r_{i+1} = \delta(s_{i+1}, w_{i+1}) = \delta(\delta(r_i, z_{i+1}), w_{i+1})$$

Therefore $r_{i+1} \in \eta(r_i, w_{i+1})$ by definition of η (choosing $\alpha = z_{i+1}$). So it is a valid accepting run and we have $w \in L(\mathcal{B})$.

- Let $w = w_1 \cdots w_n \in L(\mathcal{B})$. Let $r_0, \dots, r_n$ be an accepting run of $\mathcal{B}$ on w . For every $0 \leq i < n$, we have $r_{i+1} \in \eta(r_i, w_{i+1})$. Hence there exists a letter $z_{i+1} \in \Sigma$ such that $r_{i+1} = \delta(\delta(r_i, z_{i+1}), w_{i+1})$. Denote $s_{i+1} = \delta(r_i, z_{i+1})$.

We claim that $r_0, s_1, r_1, s_2, r_2, \dots, s_n, r_n$ is an accepting run of $\mathcal{A}$ on $z_1 w_1 \cdots z_n w_n$ (hence $w \in L_{\text{mix}}$). Indeed, we have $r_0 \in \{q_0\}$ so r_0 is the initial state, and $r_n \in F$ is an accepting state. For every $0 \leq i < n$:

$$\begin{aligned} r_{i+1} &= \delta(\delta(r_i, z_{i+1}), w_{i+1}) = \delta(s_{i+1}, w_{i+1}) \\ s_{i+1} &= \delta(r_i, z_{i+1}) \end{aligned}$$

So it is a valid accepting run. This choice of z shows that $w \in L_{\text{mix}}$.

Hence $L_{\text{mix}} \in \text{NREG}$.

2 Determinization

We now show how to construct a DFA that is equivalent to a given NFA and thus obtain that $\text{REG} = \text{NREG}$.

2.1 Subset construction

Let $\mathcal{A} = \langle Q, \Sigma, \delta, Q_0, F \rangle$, be an NFA. We define an equivalent DFA $\mathcal{B} = \langle P, \Sigma, \alpha, p_0, G \rangle$ as follows:

- Every state in the DFA is a subset of states in the NFA: $P = 2^Q$.
- The alphabet Σ is unchanged.
- The accepting states are those that intersect F : $G = \{S \subseteq Q : S \cap F \neq \emptyset\}$.
- The initial state of the DFA is the set of initial states in the NFA: $p_0 = Q_0$.
- The transition function includes every state in the NFA reachable from here: for every $S \subseteq Q$, define $\alpha(S, \sigma) = \bigcup_{q \in S} \delta(q, \sigma)$.

Proposition 1. *The NFA $\mathcal{A}$ and DFA $\mathcal{B}$ are equivalent, i.e. $L(\mathcal{A}) = L(\mathcal{B})$.*

Proof. First, we show that for every $w \in \Sigma^*$ and $S \subseteq Q$, we have $\alpha^*(S, w) = \delta^*(S, w)$. That is, δ^* and α^* are the same function (but remember that δ^* belongs to an NFA and α^* belongs to a DFA). The proof is by induction on $|w|$.

- **Base case:** If $|w| = 0$ then $w = \epsilon$, and by definition we have $\alpha^*(S, \epsilon) = S = \delta^*(S, \epsilon)$.
- **Step:** Assume that the claim is correct for some $w \in \Sigma^*$. For a longer word $w\sigma$ ($\sigma \in \Sigma$) we have:

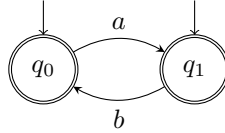
$$\alpha^*(S, w\sigma) \stackrel{(1)}{=} \alpha(\alpha^*(S, w), \sigma) \stackrel{(2)}{=} \alpha(\delta^*(S, w), \sigma) \stackrel{(3)}{=} \bigcup_{q \in \delta^*(S, w)} \delta(q, \sigma) \stackrel{(4)}{=} \delta^*(S, w\sigma)$$

where (1) is by definition of α^* , (2) is by the induction hypothesis, (3) is by definition of α , and (4) is by definition of δ^* .

Now $w \in L(\mathcal{A})$ iff $\delta^*(Q_0, w) \cap F \neq \emptyset$ iff $\alpha^*(Q_0, w) \cap F \neq \emptyset$ iff $\alpha^*(Q_0, w) \in G$ iff $w \in L(\mathcal{B})$, and we are done. $\square$

2.2 Example

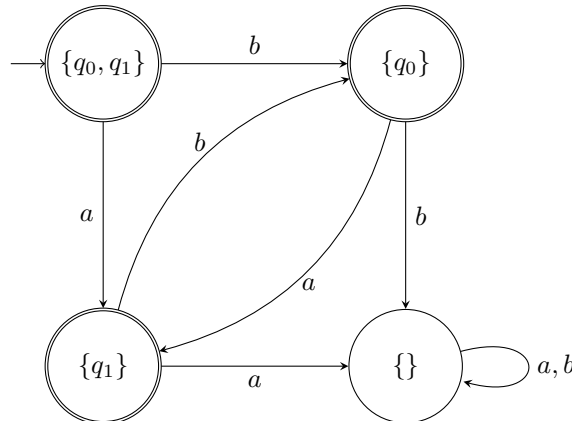
Let's look at the following NFA:



Intuitively, we see that we are allowed to alternate between a and b . If we see aa or bb at any point, the run cannot continue. We can phrase it as follows:

$$L(\mathcal{A}) = \{w \subseteq \Sigma^* : w \text{ does not contain two identical consecutive letters}\}$$

After subset-construction we get:



2.3 Determine Acceptance in an NFA

How do we determine acceptance of a word $w = \sigma_1 \cdot \dots \cdot \sigma_l$ over an NFA with n states? We show three different options.

2.3.1 Determinization and running of the DFA

Time complexity - $O(2^n \cdot |\Sigma|)$ for the determinization + $O(|w|)$ for running the DFA.

Space complexity - $O(2^n \cdot |\Sigma|)$ for the DFA.

2.3.2 Brute Force

Time complexity - $O(n^{|w|})$ ($O(k^{|w|})$ if there are at most k splits per state) to go over all possible runs of the NFA over w .

2.3.3 Subset Construction Simulation

Subset Construction Simulation - Iteratively evaluate all possible transition states for each letter $\sigma_1, \dots, \sigma_l$ of w . Accept if the final set contains an accepting state.

Algorithm 1 Subset Construction Simulation

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1:  $S \leftarrow Q_0$ 
2: for  $i = 1$  in  $\{1, \dots, |w|\}$  do
3:    $S \leftarrow \bigcup_{q \in S} \delta(q, \sigma_i)$ 
4: end for
5: if  $F \cap S \neq \emptyset$  then
6:   accept
7: else
8:   reject
9: end if

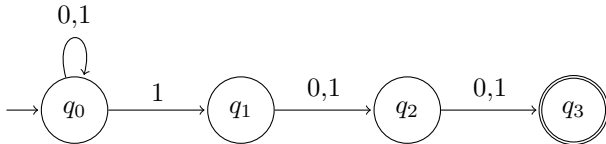
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Proposition 2. After every step i , $S = \delta^*(Q_0, w_{\leq i})$, where $w_{\leq i} = w_1 \dots w_i$.

The proof is by induction over $|w|$ and left as an exercise.

Time complexity - $|w|$ iterations, in each iteration we evaluate δ for at most n states, each evaluation may lead to n states and hence the complexity is $O(n^2 \cdot |w|)$.

Example L_k for $k = 3$



We simulate the run over the word $w = 011000$:

1. $S = Q_0 = \{q_0\}$
2. $S = \delta(q_0, 0) = \{q_0\}$
3. $S = \delta(q_0, 1) = \{q_0, q_1\}$
4. $S = \delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0, q_1, q_2\}$
5. $S = \delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0) = \{q_0, q_2, q_3\}$
6. $S = \delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_3, 0) = \{q_0, q_3\}$
7. $S = \delta(q_0, 0) \cup \delta(q_3, 0) = \{q_0\}$

$F \cap S = \emptyset \rightarrow w$ is not accepted

3 Myhill-Nerode equivalence

3.1 Separating suffix

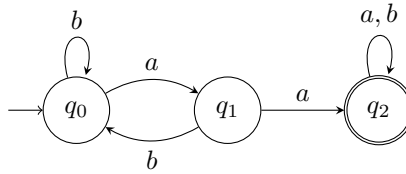
Definition 3 (Separating Suffix). Let $L \subseteq \Sigma^*$ be a language and let $x, y \in \Sigma^*$ be words. We say that $z \in \Sigma^*$ is a separating suffix for x and y with respect to L if $x \cdot z \in L$ and $y \cdot z \notin L$ or vice versa.

Example: In a previous recitations we saw the language

$$L = \{w \in \{a, b\}^* : w \text{ contains } aa\}.$$

- a is a separating suffix for $bbba$ and b have a separating suffix, since $bbbaa \in L$ and $ba \notin L$.
- $babab$ and ϵ do not have a separating suffix.

We also saw a DFA that decides this language.



Let's formally prove an idea we already saw in the last recitations.

Proposition 4. Let $L \in \text{REG}$ and let $\mathcal{A} = \langle Q, \Sigma, \delta, q_0, F \rangle$ be a DFA that recognizes L . Suppose that $x, y \in \Sigma^*$ are words that reach the same state in $\mathcal{A}$, that is, $\delta^*(q_0, x) = \delta^*(q_0, y)$. Then x, y do not have a separating suffix.

Proof. Recall from the previous recitation that $\delta^*(q_0, ww') = \delta^*(\delta^*(q_0, w), w')$ for every $w, w' \in \Sigma^*$. Let $z \in \Sigma^*$. We have:

$$\delta^*(q_0, xz) = \delta^*(\delta^*(q_0, x), z) = \delta^*(\delta^*(q_0, y), z) = \delta^*(q_0, yz)$$

Therefore xz, yz reach the same state. If that state is accepting, both belong to L . Otherwise, both do not belong to L . $\square$

The contrapositive of the above proposition is: if x, y have a separating suffix, then they reach different states in $\mathcal{A}$. We will see how to use this proposition to show a lower bound on the number of states in a DFA.

Going back to our example, we can now prove that there is no smaller DFA that recognizes L . Consider the words a, b, aa . Every pair of them has a separating suffix:

- a, b have a separating suffix a , since $aa \in L$ and $ba \notin L$.
- a, aa have a separating suffix ϵ , since $a \notin L$ and $aa \in L$.
- b, aa have a separating suffix ϵ , since $b \notin L$ and $aa \in L$.

We conclude that in every DFA $\mathcal{A}$ that recognizes L , each of the words a, b, aa reaches a different state, so $\mathcal{A}$ has at least 3 states.

Therefore, the DFA we saw in the recitation is minimal.

3.2 Equivalence relation

Definition 5. Let $L \subseteq \Sigma^*$. Two words $x, y \in \Sigma^*$ are called Myhill-Nerode equivalent with respect to L , denoted $x \sim_L y$, if they do not have a separating suffix. That is, for every $z \in \Sigma^*$, we have $xz \in L \Leftrightarrow yz \in L$.

This is an equivalence relation:

- Reflexivity: $x \sim_L x$.
- Symmetry: if $x \sim_L y$ then $y \sim_L x$.
- Transitivity: if $x \sim_L y$ and $y \sim_L w$ then $x \sim_L w$.

The full proof is left as an exercise. Let us show transitivity: suppose that $x \sim_L y$ and $y \sim_L w$. Then for every $z \in \Sigma^*$ we have:

$$xz \in L \Leftrightarrow yz \in L \Leftrightarrow wz \in L$$

So $xz \in L \Leftrightarrow wz \in L$, hence $x \sim_L w$.

Example: Going back to our example, $L = \{w : w \text{ contains } aa\}$ has 3 different Myhill-Nerode equivalence classes:

1. Words that contain aa .
2. Words that don't contain aa , and end in a .
3. Words that don't contain aa , and don't end in a .

Let x, y be words in the first set. Since they contain aa , then xz, yz also contain aa for every $z \in \Sigma^*$. Hence $xz, yz \in L$, so $x \sim_L y$. Similarly, for each of the other two sets, it can be shown that every two words in the set are equivalent.

We can also show that each of the three sets is a distinct equivalence class, by using a separating suffix. Let x be a word from the second set, and let y be a word from the third. Then $z = a$ is a separating suffix: since x ends in a , the word xz contains aa so it is in L . But yz does not contain aa , so it is not in L . The other cases are similar.

Note that each equivalence class correspond to a state in the DFA we constructed and contain the words that end their run in this state. Is it a coincidence?

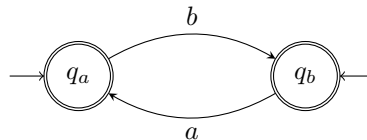
Proposition 6. *Given a regular language L , each Myhill-Nerode equivalence class of L corresponds to a state in the minimal DFA that decides L . In particular, the number of states in the minimal DFA is the number of Myhill-Nerode equivalence classes.*

We saw the proof in the lecture.

Example: We previously saw the language

$$L = \{w \in \{a, b\}^* : w \text{ does not contain two identical consecutive letters}\}$$

We described an NFA for L with 2 states, and after determinizing it we obtained a DFA with 4 states.



We now show that 4 states are indeed necessary in this case, by listing separating suffixes for each pair of the words ϵ, a, b, aa .

- ϵ, a have a separating suffix a , since $a \in L$ and $aa \notin L$.
- ϵ, b have a separating suffix b , since $b \in L$ and $bb \notin L$.
- ϵ, aa have a separating suffix ϵ , since $\epsilon \in L$ and $aa \notin L$.
- a, b have a separating suffix b , since $ab \in L$ and $bb \notin L$.
- a, aa have a separating suffix ϵ , since $a \in L$ and $aa \notin L$.
- b, aa have a separating suffix ϵ , since $b \in L$ and $aa \notin L$.

The equivalence *Myhill-Nerode* classes are as follows:

- $[\epsilon]$ - The set that contains only the empty word.
- $[a]$ - the words that end in a .
- $[b]$ - the words that end in b .
- $[aa]$ - the words that contain a repeated letter.

We can also use separating suffixes to show that a language is not regular:

Proposition 7. *The language $L = \{w \in \{a, b\}^* : \#_a(w) \geq \#_b(w)\}$ is not regular.*

Proof. Suppose for the sake of contradiction that there is a DFA $\mathcal{A} = \langle Q, \Sigma, \delta, q_0, F \rangle$ that recognizes L . Let $k = |Q|$. Consider the following words:

$$\epsilon, a, aa, aaa, \dots, a^k$$

There are $k + 1$ words in total, so by the pigeonhole principle, two of them reach the same state. Let a^i, a^j be such a pair where without loss of generality $i > j$. b^i is a separating suffix for a^i and a^j , contradicting the proposition we saw. $\square$

Example (if time permits): $L = \{w : |w| \geq 3\}$ over $\Sigma = \{a, b\}$. There are 4 different equivalence classes:

1. $|w| = 0$
2. $|w| = 1$
3. $|w| = 2$
4. $|w| \geq 3$

We show that the classes are distinct. Let x, y be words such that $0 \leq |x| < |y| \leq 3$. Then $z = a^{3-|y|}$ is a separating suffix: we have this was just for $|y| \leq 3$

$$|yz| = |y| + |z| = |y| + |a^{3-|y|}| = |y| + 3 - |y| = 3$$

so $yz \in L$. On the other hand,

$$|xz| = |x| + |z| = |x| + |a^{3-|y|}| = |x| + 3 - |y| < 3$$

so $xz \notin L$.

By Proposition 4, if L is regular then the number of Myhill-Nerode equivalence class is a lower bound on the number of states in the DFA.

Proposition 8. *The language $L = \{1^{n^2} : n \geq 0\}$ is not regular.*

Example: We show L has infinitely many Myhill-Nerode equivalence classes, so it is not regular. In particular, we show that for every $0 \leq i < j$, the words $x = 1^{i^2}$ and $y = 1^{j^2}$ belong to distinct classes. Let $z = 1^{2i+1}$. We have that

$$|xz| = |x| + |z| = i^2 + 2i + 1 = (i + 1)^2,$$

so $xz \in L$. On the other hand,

$$|yz| = |y| + |z| = j^2 + 2i + 1 < j^2 + 2j + 1 = (j + 1)^2.$$

We have $j^2 < |yz| < (j + 1)^2$. Its length is not a square number, so $yz \notin L$.

Proposition 9. *The language $L = \{1^p : p \text{ is a prime number}\}$ is not regular.*

We give a sketch of a proof.

Every infinite **unary regular** language contains an arithmetic sequence. As there is a finite number of states, it must be that there is a loop of size k . If $w \in L$, then also $w \cdot \underbrace{1 \dots 1}_k \in L$, and $w \cdot \underbrace{1 \dots 1}_{2k}$, and so on.

The primal numbers do not contain an arithmetic sequence. $\forall n \in \mathbb{N}$, there is a sequence of size $n - 1$ which is not primal. We look at $n!$, and $n! + 2, n! + 3, \dots, n! + n$. Since $n!$ is not primal, and is a multiplication of 2 by definition, then $n! + 2$ is not primal as it is divisible by 2. Similarly, $n! + 3$ is not primal as it is divisible by 3, and so on.